

3 (**short**) Figure 3 shows a MOSFET which is to be biased to an operating point of $V_{GS} = 4\text{ V}$, $V_{DS} = 10\text{ V}$ and $I_{DS} = 4\text{ mA}$.

(a) Calculate the values of R_1 and R_2 to achieve this. [5]

(b) If $r_d = 30\text{ k}\Omega$ and $g_m = 5\text{ mS}$ for the MOSFET, calculate the mid-band small signal gain of the circuit. [5]

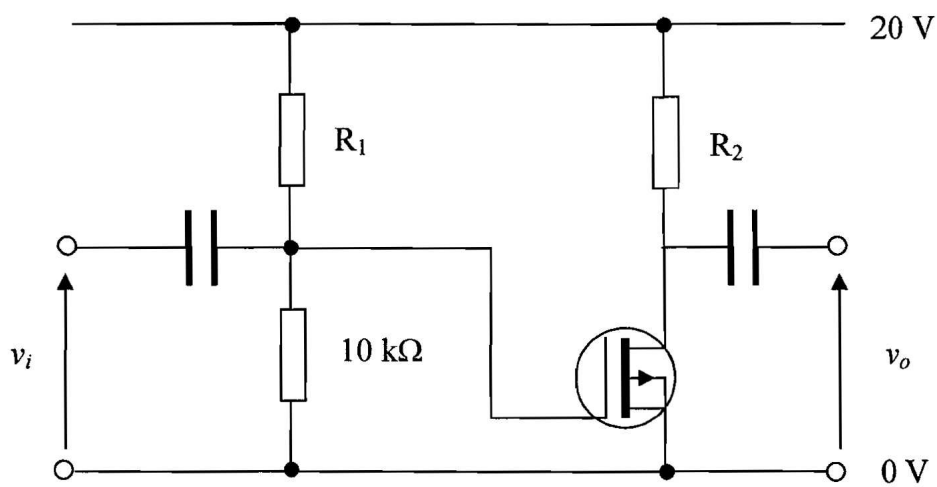


Fig. 3